

Light or Full Verb?

A Minimal-Pair Dataset for Probing Phraseological Competence in Language Models

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Light verbs, full verbs:

The same verb can occur in a **light verb construction** (*make a decision*) or as a **full lexical verb** (*make a cake*). In these constructions the verb looks alike, while its lexical and semantic properties differ.

Light verb: The verb contributes little lexical meaning; the noun carries the main semantic content (*decision* denotes the act of deciding).

Full verb: The verb retains its lexical meaning and contributes centrally to the event described (*make* denotes creating something).

- RQ1:** Do LMs assign different probabilities to light- and full-verb constructions?
RQ2: Is this distinction reflected in contextual embeddings?
RQ3: How consistent are these patterns across languages and models?

Surprisal
Surprisal-Based Probing
Does light-verb status affect word predictability?
Surprisal measures how unexpected a word is in context:
[
S(w_t)=-\log P(w_t\mid w_{<t})
]
Active: Measure surprisal of the object noun following the verb.
Passive: Measure surprisal of the verb following the nominal base.
Results
LVCs generally have **lower surprisal** than matched full-verb constructions across six tested language models and three languages (Holm-adjusted $p \leq .033$).
The effect varies by verb, with *receive* and its equivalents showing less consistent patterns.

Dataset

Dataset

We introduce a controlled multilingual dataset contrasting **light-verb constructions (LVCs)** and **full-verb constructions (FVCs)** within matched sentence contexts.
3 languages: English · French · Spanish
5 verbs per language: make, take, give, have, receive and their equivalents.
Matched sentence examples

Light verb (LV)

Full verb

Active

They made a **decision**.

They made a **cake**.

Passive

A **decision** was made.

A **cake** was made.

Dataset construction

1. Select from word lists and validated predicative nouns.
2. Pair them with natural full-verb + concrete-object combinations.
3. Generate matched sentence contexts varying voice, tense, and other grammatical properties.
Experimental scale
•47,600 English sentences per voice
•41,310 French sentences per voice
•45,080 Spanish sentences per voice
All examples are grammatical; the contrast concerns the **lexical-functional status** of the verb.

Embeddings

Contextual Embedding Analysis

Do LVCs and FVCs occupy different regions of representation space?

We extract contextual representations of object noun phrases and compare them using dimensionality reduction and clustering.

Results
LVC and full-verb objects show clear separation across many models and layers. Separation is more stable in accumulated representations than in layer-to-layer changes.

The strength of separation varies across models and languages.

Key observation: The lexical-functional contrast is reflected in contextual representations, although noun semantics also contributes.